Ring and case: every conductor, one line each (from design/wiring/product-wiring.csv, 67 lines)

drawn and routed · specified and landed · drawn, not routed · stated in text only · not drawn

R3 joystick switch lines (FLEX1) (ends also in R1) 10 lines

FLEX1	BTN_REC (via land J_BAND_A:5)	SW2.3 BTN_REC	FLEX1 SW2
	W010 FLEX1 13.302-15.235 mm		
off board SW1, SW3, SW4, SW5	SW1.1 JOY_UP	SW1.3 JOY_UP	off board
	W006 inside the switch		
	SW3.1 JOY_DOWN	SW3.3 JOY_DOWN	
	W007 inside the switch		
	SW4.1 JOY_LEFT	SW4.3 JOY_LEFT	
	W008 inside the switch		
	SW5.1 JOY_RIGHT	SW5.3 JOY_RIGHT	
	W009 inside the switch		
ring main board J_BAND_A	J_BAND_A.1 JOY_UP	SW1.3 JOY_UP	FLEX1 SW1, SW3, SW4, SW5, SW2
	W001 FLEX1 11.577-13.519 mm		
	J_BAND_A.2 JOY_DOWN	SW3.3 JOY_DOWN	
	W002 FLEX1 20.219-22.164 mm		
	J_BAND_A.3 JOY_LEFT	SW4.3 JOY_LEFT	
	W003 FLEX1 14.569-16.513 mm		
	J_BAND_A.4 JOY_RIGHT	SW5.3 JOY_RIGHT	
	W004 FLEX1 15.815-17.758 mm		
	J_BAND_A.5 BTN_REC	SW2.1 BTN_REC	
	W005 FLEX1 12.552-14.485 mm		

R4 joystick switch grounds (FLEX1) (ends also in R1) 10 lines

FLEX1	GND (via land J_BAND_A:6)	SW1.4 GND	FLEX1 SW1, SW2, SW3, SW5
	W016 FLEX1 18.385-20.328 mm		
	GND (via land J_BAND_A:6)	SW2.2 GND	
	W017 FLEX1 16.735-18.678 mm		
	GND (via land J_BAND_A:6)	SW2.4 GND	
	W018 FLEX1 16.735-18.678 mm		
	GND (via land J_BAND_A:6)	SW3.4 GND	
	W019 FLEX1 17.985-19.928 mm		
	GND (via land J_BAND_A:6)	SW5.4 GND	
	W020 FLEX1 18.385-20.328 mm		
off board SW1, SW3, SW4, SW5	SW1.2 GND	SW1.4 GND	off board
	W012 inside the switch		
	SW3.2 GND	SW3.4 GND	
	W013 inside the switch		
	SW4.2 GND	SW4.4 GND	
	W014 inside the switch		
	SW5.2 GND	SW5.4 GND	
	W015 inside the switch		
ring main board J_BAND_A	J_BAND_A.8 GND	SW4.4 GND	FLEX1 SW4
	W011 FLEX1 11.685-13.628 mm		

R5 motor M1: FLEX1 lines and motor leads (ends also in R1) 4 lines

motor M1 M1	M1.1 VMOT	M1.1 VMOT	FLEX1 M1
	W023 red lead wire AWG32 of the motor, cut 1.07-1.44 mm		
	M1.2 GND	M1.2 GND	
	W024 blue lead wire AWG32 of the motor, cut 1.07-1.44 mm		
ring main board J_BAND_A	J_BAND_A.6 GND	M1.2 GND	
	W021 FLEX1 7.786-8.980 mm		
	J_BAND_A.7 VMOT	M1.1 VMOT	
	W022 FLEX1 5.737-6.931 mm		

R6 cell lines on FLEX2 (to the PRT1 tab lands) (ends also in R2) 2 lines

ring main board J_BAND_B	J_BAND_B.1 VBAT	PRT1.1 VBAT	FLEX2 PRT1
	W025 FLEX2 5.995-6.799 mm		
	J_BAND_B.2 GND	PRT1.2 GND	
	W026 FLEX2 7.233-8.037 mm		

R7 thermistor line on FLEX2 (ends also in R2) 1 line

ring main board J_BAND_B	J_BAND_B.3 NTC	PRT1.3 NTC	FLEX2 PRT1
	W027 FLEX2 6.505-7.310 mm		

R8 thermistor ground on the PRT1 board (ends also in R10) 1 line

cell pack BT1 + PRT1 + RT1, one purchased part RT1	RT1.2 GND	BT1.2 GND	inside the pack
	W028 inside the cell pack		

R9 charging pads CP1 to CP4 on FLEX2 (ends also in R2) 4 lines

ring main board J_BAND_B	J_BAND_B.4 VBUS	CP1.1 VBUS	FLEX2 CP1, CP2, CP3, CP4
	W029 FLEX2 23.182-30.514 mm		
	J_BAND_B.5 USB_DN	CP2.1 USB_DN	
	W030 FLEX2 25.946-33.269 mm		
	J_BAND_B.6 USB_DP	CP3.1 USB_DP	
	W031 FLEX2 28.704-36.018 mm		
	J_BAND_B.7 GND	CP4.1 GND	
	W032 FLEX2 32.860-40.165 mm		

R10 cell protection circuit PRT1: output tabs (ends also in R6, R7) 3 lines

cell pack BT1 + PRT1 + RT1, one purchased part BT1, RT1	BT1.1 VBAT	PRT1.1 VBAT	FLEX2 PRT1
	W033 PRT1 tab (nickel strip)		
	BT1.2 GND	PRT1.2 GND	
	W034 PRT1 tab (nickel strip)		
	RT1.1 NTC	PRT1.3 NTC	
	W035 PRT1 tab (nickel strip)		

R11 antenna feed (main board copper) 1 line

ring main board ANT1	ANT1.1 RF_FEED	RF_FEED (C13:1 L4:2)	ring main board
	W036 main board copper		

R12 antenna ground and trim terminals 3 lines

ring main board ANT1	ANT1.2 ANT_TRIM	ANT_TRIM (LT1:1)	ring main board
	W037 main board copper		
	ANT1.3 ANT_TRIM	ANT_TRIM (LT1:1)	
	W038 main board copper		
	ANT1.4 GND	GND (ground)	
	W039 main board copper		

R13 screen cable into J1 14 lines

screen DISP1 DISP1	DISP1.1 ELVSS	J1.1 ELVSS	ring main board J1
	W040 screen's own cable, contact 1		
	DISP1.10 LCD_SCK	J1.10 LCD_SCK	
	W041 screen's own cable, contact 10		
	DISP1.11 LCD_DCX	J1.11 LCD_DCX	
	W042 screen's own cable, contact 11		
	DISP1.12 LCD_MOSI	J1.12 LCD_MOSI	
	W043 screen's own cable, contact 12		
	DISP1.13 GND	J1.13 GND	
	W044 screen's own cable, contact 13		
	DISP1.14 GND	J1.14 GND	
	W045 screen's own cable, contact 14		
	DISP1.15 GND	J1.15 GND	
	W046 screen's own cable, contact 15		
	DISP1.2 ELVDD	J1.2 ELVDD	
	W047 screen's own cable, contact 2		
	DISP1.4 VDD_3V0	J1.4 VDD_3V0	
	W048 screen's own cable, contact 4		
	DISP1.5 VDD_1V8	J1.5 VDD_1V8	
	W049 screen's own cable, contact 5		
	DISP1.6 SWIRE	J1.6 SWIRE	
	W050 screen's own cable, contact 6		
	DISP1.7 LCD_TE	J1.7 LCD_TE	
	W051 screen's own cable, contact 7		
	DISP1.8 LCD_RST	J1.8 LCD_RST	
	W052 screen's own cable, contact 8		
	DISP1.9 LCD_CS	J1.9 LCD_CS	
	W053 screen's own cable, contact 9		

R15 ring pads to case spring pins P1 to P4 (ends also in C4, R9) 4 lines

FLEX2 CP1, CP2, CP3, CP4	CP1.1 VBUS	P1.1 VRING	case pin board PCB2 P1, P2, P3, P4
	W054 spring pin (pressed)		
	CP2.1 USB_DN	P2.1 USB_D-	
	W055 spring pin (pressed)		
	CP3.1 USB_DP	P3.1 USB_D+	
	W056 spring pin (pressed)		
	CP4.1 GND	P4.1 GND	
	W057 spring pin (pressed)		

C1 case cell lead + 1 line

case cell BT1 BT1	BT1.1 BAT	W5.1 BAT	case main board PCB1 W5
	W059 cell's own flying lead (red +), cut 13.0 mm		

C2 case cell lead - 1 line

case cell BT1 BT1	BT1.2 GND	W6.1 GND	case main board PCB1 W6
	W060 cell's own flying lead (black -), cut 15.2 mm		

C3 case thermistor TH1 leads 2 lines

thermistor on the cell face TH1	TH1.1 TS	TH1.1 TS	case main board PCB1 TH1
	W061 thermistor's own lead, bare, no polarity, cut 10 mm		
	TH1.2 GND	TH1.2 GND	
	W062 thermistor's own lead, bare, no polarity, cut 10 mm		

C5 case wires pin board to main board 5 lines

case pin board PCB2 WP1, WP2, WP3, WP4, WP7	WP1.1 VRING	W1.1 VRING	case main board PCB1 W1, W2, W3, W4, W7
	W063 wire 30 AWG stranded, red insulation, cut 15.6 mm		
	WP2.1 USB_D-	W2.1 USB_D-	
	W064 wire 30 AWG stranded, white insulation, cut 16.3 mm		
	WP3.1 USB_D+	W3.1 USB_D+	
	W065 wire 30 AWG stranded, green insulation, cut 16.3 mm		
	WP4.1 GND	W4.1 GND	
	W066 wire 30 AWG stranded, black insulation, cut 15.6 mm		
	WP7.1 RING_TOUCH	W7.1 RING_TOUCH	
	W067 wire 30 AWG stranded, yellow insulation, cut 15.9 mm		

SENSING GAP, NO CONDUCTOR (counted apart from the conductors): R16 band metal to case touch electrode E1 (no conductor) 1 line

ring band metal, no file holds it as an item	band metal (no terminal)	E1.1 RING_TOUCH	case pin board PCB2 E1
	W058 no conductor, gap 2.04-2.20 mm		

Line style = state of the line in the csv (column state). Copper lengths are the shortest copper path pad to pad in the band boards (smallest to largest over US 7 to 13); cut lengths are the lead or wire cut length (smallest to largest over US 7 to 13). Made by design/wiring/draw_wiring.py.